The empirical recurrence for OEIS A303459, recovered from its tail

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Abstract

OEIS A303459 records "Empirical recurrence of order 68" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 124$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{68} = 26052460544 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303459 is "Number of 5Xn 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$16, 512, 12408, 306767, 7626768, 189848373, 4729712941, 117851999934, \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Apr 24 2018 10:24:59, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 124$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 124$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 248$ exact terms from $n = 2$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{68} c_i \, a(n-i),$$

c_1	30	c_{18}	−457506049517	c_{35}	−42738628088261859	c_{52}	147106762330116315
c_2	−6	c_{19}	2485131288959	c_{36}	23647292475638405	c_{53}	36595398946940118
c_3	−3543	c_{20}	−3169312436783	c_{37}	−36638008573877058	c_{54}	−153805305928561960
c_4	8976	c_{21}	−11296185293449	c_{38}	218632698074320348	c_{55}	−57446065898951826
c_5	130068	c_{22}	19647421619330	c_{39}	154590847993975711	c_{56}	−257821355703474800
c_6	−442236	c_{23}	−54479187454252	c_{40}	−163385286501961639	c_{57}	−1856367685271028
c_7	−1294336	c_{24}	111397592842131	c_{41}	−87529091535405116	c_{58}	4829887561176532
c_8	6186197	c_{25}	424306895988084	c_{42}	−257382330437260024	c_{59}	3862871959554840
c_9	−19696885	c_{26}	−737457494288530	c_{43}	−111873072312665628	c_{60}	577095590949664
c_{10}	44117316	c_{27}	43471265332443	c_{44}	318221675741786391	c_{61}	−55744253763520
c_{11}	396950548	c_{28}	−1198655186698842	c_{45}	249479164798419927	c_{62}	−113365155462144
c_{12}	−1628517238	c_{29}	−4859334907106118	c_{46}	103066838503449768	c_{63}	−57240216132992
c_{13}	238183276	c_{30}	11075509449399870	c_{47}	−195374008359610204	c_{64}	−10617657520128
c_{14}	7510478631	c_{31}	5014787879755939	c_{48}	−325325965822929086	c_{65}	972564246528
c_{15}	−36270211800	c_{32}	3516362146324244	c_{49}	−128416695992477862	c_{66}	127111917568
c_{16}	71520698840	c_{33}	22605793809234300	c_{50}	20068547044199044	c_{67}	118295052288
c_{17}	50347589721	c_{34}	−74070967420036176	c_{51}	185748116222762832	c_{68}	26052460544

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A303459 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{68-1} - \cdots - c_{68}$. Its constant term is $-c_{68} = -26052460544$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 26052460544.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A303459.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.